Practice workbook · Finding a curve from its gradient function

Integrating dy/dx term by term, then using given points (and sometimes the factor theorem) to find the constant and any unknowns.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-CAL-16

1. Question 7 · part a.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part a.i only.

7 The point with coordinates $(4, -104)$ lies on the curve C with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that C passes through the origin,

(4)

(ii) show that C has a maximum at the point on the curve where $x = 0.5$

(3)

The curve C has another turning point at A and another turning point at B.

Given that the x coordinate of A is negative,

(b) (i) find the coordinates of A and the coordinates of B,

(5)

(ii) determine the nature of these turning points.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20

1. Question 7 · part a.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part a · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

9 The curve C, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that C passes through the origin.

(4)

(b) (i) Show that C has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the y coordinate at each of these points.

(7)

The curve has another turning point at A.

(c) (i) Find the coordinates of A.

(ii) Determine the nature of this turning point.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 9 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part whole · 9 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

8 $f'(x) = 18x^2 - 2x + 13$

Given that $(2x - 1)$ is a factor of $f(x)$

show that the curve with equation $y = f(x)$ has only one intersection with the x-axis.

(9)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 8 · part whole · 9 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 8 continued

OLevels.com SCHOOL

(Total for Question 8 is 9 marks)

3. Question 8 · part whole · 9 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part a · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

10 The equation of a curve C is $y = f(x)$ where $f'(x) = 3x^2 - 4x - p$ and $p \neq 0$

The points with coordinates $(2, 0)$ and $(-1, 9)$ lie on C.

(a) Show that C has equation $y = x^3 - 2x^2 - 4x + 8$

(6)

The straight line l has equation $y = 8 - 4x$

(b) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 10 · part a · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part a · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

8

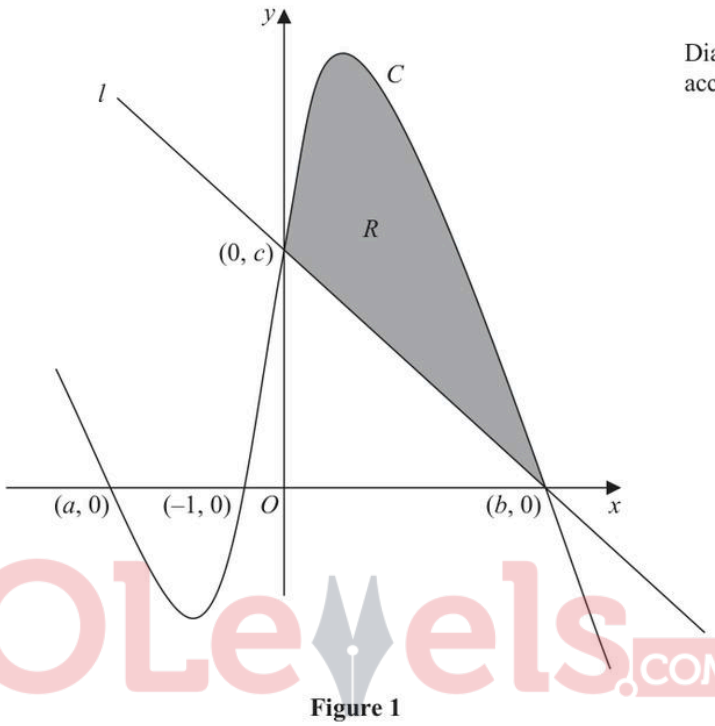


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation $y = f(x)$

The curve C passes through the points with coordinates

$(a, 0)$, $(-1, 0)$, $(b, 0)$ and $(0, c)$

Given that $f'(x) = 17 + 2x - 3x^2$

(a) show that the equation of C is $y = 15 + 17x + x^2 - x^3$

(4)

(b) Find the value of a, the value of b and the value of c

(6)

The straight line l intersects C at the points with coordinates $(b, 0)$ and $(0, c)$

The region R, shown shaded in Figure 1, is bounded by l and C

(c) Use algebraic integration to find the exact area of region R

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

P 7 1 8 1 8 A 0 1 8 2 8

5. Question 8 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part a · 5 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

3 $g'(x) = mx^2 - 10x - 37$ where m is an integer

The curve $y = g(x)$ passes through the point with coordinates $(1, 20)$

Given that $(x - 5)$ is a factor of $g(x)$

(a) show that $g(x) = 2x^3 - 5x^2 - 37x + 60$

(5)

(b) Hence, or otherwise, use algebra to solve the equation $g(x) = 0$

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 3 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL